Replacement of provably equal terms

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stub ax:id2; note prop:iden1 (source lines 762-764) has no proof environment of its own in the source and is not a counted card - the census pairing artifact is recorded here.

Editorial note. The label prop:iden2 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Identity axiom scheme ax:id2. Same unit, \ollabel{ax:id2} (identity axiom group, source line 746). The displayed formula is an instance of the scheme.

Source reference. Original reference ax-id2: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 744–744. Prerequisite pointer invoked at proof line(s) 776; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *If* $\Gamma \vdash A(t_1)$ *and* $\Gamma \vdash t_1 = t_2$*, then* $\Gamma \vdash A(t_2)$.

Proof. The formula

$$(t_1 = t_2 \to (A(t_1) \to A(t_2)))$$

is an instance of ax-id2. The conclusion follows by two applications of MP. □